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An Analytical Technique for Modeling Knee Joint Stiffness— Part I: Ligamentous Forces¹

The purpose of this study is to present an analytical technique to evaluate the nonlinear, coupled stiffness characteristics of the human knee joint. In the present work, a model of the knee is developed which permits the contribution of the ligaments to the nonlinear, coupled stiffness characteristics of the knee to be determined from their nonlinear load-elongation properties, the geometric location of their insertion sites, and the relative position of the femur with respect to the tibia. Calculations with the model are compared to those obtained using Crowninshield, et al. [1] model and experimental data.

Introduction

It has become increasingly popular to use the methods of structural mechanics in constructing two-dimensional and three-dimensional mathematical models of biomechanical structures, including those in the human body. Some investigators have used the familiar structural techniques [1–4] as their tools while others preferred utilizing the more recent matrix structural analysis and finite element techniques [5–9]. Belytschko, et al. [5] developed a mathematical model for a three-dimensional force analysis of the human vertebral column where an incremental stiffness method which accounts for nonlinearities due to large displacements was used. Andriacchi, et al. [6] used the computational techniques described by Belytschko, et al. to develop a model useful for studies of the mechanics of the human skeletal thorax. Panjabi [7] presented a method for constructing three-dimensional mathematical models and governing equations of motion of the spine. Hayes, et al. [8] and others have used finite element codes to predict stresses and displacements in the proximal tibia of the human knee, while Huston and Passerello [9] and others have developed mathematical models for the prediction of human body dynamics during gait, impact, and other conditions.

There are relatively few published analytical models of the knee, probably due to its complexity and the lack of pertinent experimental data. In studies of whole human body dynamics, the knee is usually considered to be a simple pin joint with no detail given to either joint geometry or ligament function.

Morrison [10] employed the combined experimental and modeling methods developed by Paul [11] to study the forces transmitted by the knee during walking. In order to determine how the transmitted forces were shared among the ligaments and muscles, it was necessary for Morrison to model the

functional roles of the ligaments based upon their spatial orientations. Redundancy in the equations for predicting muscle forces was eliminated using EMG data.

Piziali and Rastegar [12], Piziali, et al. [13] and Rastegar, et al. [14] used the stiffness influence coefficient method to experimentally determine the coupled nonlinear load-displacement characteristics of the human knee. These measured data were fitted with a least square cubic spline and then differentiated with respect to the imposed displacement to determine the corresponding stiffness curves.

Several models have been developed to study the relation between knee motion and ligament length changes. The methodologies used involve measuring joint kinematics and the attachment points of the ligament at some reference condition—usually full extension. The joint is displaced “mathematically” according to the specific motion, and the change in distance between attachment points calculated.

Edwards, et al. [15] compared model calculations with measurements of ligament length changes made by mercury strain gages sutured to the ligament surface. Bartel, et al. [16] determined radiographically the distance between ligament attachments for relative knee motion. Lew and Lewis [17] predicted ligament length patterns through the simulation of knee motion using an anthropometric scaling procedure. Crowninshield, et al. [1] extended Edward’s et al. model to describe 13 separate ligamentous structures and calculated joint stiffness in addition to ligament lengths. In calculating the stiffness of the knee joint, Crowninshield, et al. assumed all ligaments had the same nonlinear material properties and that they were strained to a maximum of 5 percent during passive flexion. To calculate joint stiffness Crowninshield, et al. performed a numerical displacement experiment and divided the resulting changes in joint force by the applied joint displacement. In an effort to verify the validity of their analytical model, they also computed the stiffness of the knee joint experimentally by recording the relative load-displacement characteristics for different flexion angles while displacing the tibia by known forces in varus-valgus, internal-external rotation and drawer motion.

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More extensive models of the knee have recently been proposed by Andriacchi, et al. [2], Wismans, et al. [3] and Wismans [4]. These models have been employed to predict joint motions from a known applied load. Andriacchi, et al. [2] modeled the ligaments as linear springs and noted that the knee behaved nonlinearly as a result of changes in ligament orientation. Both Wisman's, et al. and Andriacchi's, et al. models included the geometric constraints of the bones. Wisman's, et al. model, however, is restricted to the condition where both femoral condyles are in contact with the tibial plateaus.

The objective of the present study is to determine analytical expressions describing the nonlinear, coupled stiffness characteristics of the knee at a given joint position using the method of matrix structural analysis. The stiffness components are calculated in terms of the constitutive equations of the ligaments and the geometry of the joint. The geometric data include the location of the insertion sites of the ligaments and the relative position of the femur with respect to the tibia for that given joint position. The ligaments are considered here to travel in straight lines between insertion sites. Geometric nonlinearities such as the wrapping of the ligaments around each other (cruciates) and around bone surfaces are treated by Hefzy and Grood [20] in the second part of this study.

In order to demonstrate the validity of the expressions obtained for the stiffness matrix of the knee joint, computations were done using input data obtained from Crowninshield's, et al. [1]. A comparison of our stiffness results with Crowninshield's, et al. model and experimental data is also presented.

Before proceeding with the analysis, we summarize the main differences between the present approach and the one presented in [1] as follows: Crowninshield, et al. [1] obtained joint stiffnesses based upon joint displacements, while the present study calculates joint stiffnesses at a given joint position. They performed a numerical displacement experiment, calculated the changes in joint force due to this

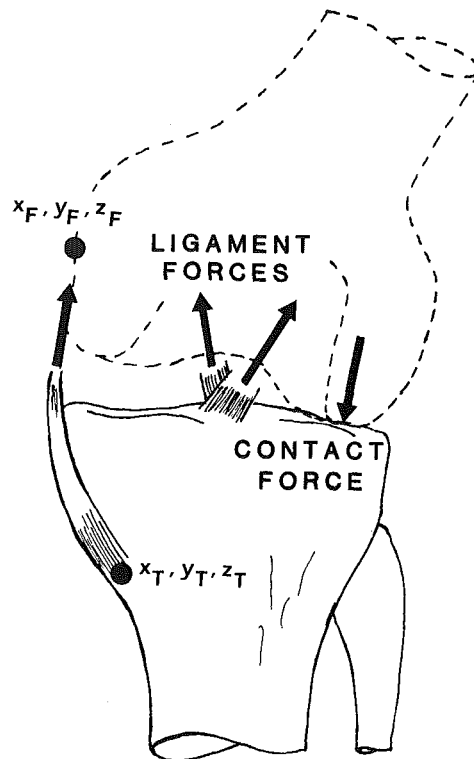


Fig. 1 Some of the forces acting on the tibia at the knee joint

displacement and divided these resulting changes in joint force by the applied joint displacement to obtain the joint stiffnesses. In contrast, in the present study we derive, in closed form, using methods of matrix structural analysis the equations which describe the nonlinear, coupled stiffness characteristics of the knee at a given joint position.

Nomenclature

$K, \bar{K}$ = secant stiffness matrices (12×12)
 $K_{11}, K_{12}, K_{21}, K_{22}$ = secant stiffness submatrices (6×6)
 R, Φ_T, Φ_F = rotation matrices (3×3)
 I = unit matrix (3×3)
 $\bar{R}$ = rotation matrix (6×6)
 S = joint tangent stiffness matrix (6×6)
 S_1, S_2, S_3, S_4 = tangent stiffness submatrices (3×3)
 L = transformation matrix (6×6)
 T = translation matrix (3×1)
 $\bar{T}$ = transformation matrix (12×12)
 W = antisymmetric position matrix (3×3)
 $\Omega(r)$ = antisymmetric tensor of vector r (3×3)

Vectors

e_i = unit vector along i th axis, $i = 1, 2, 3$
 F = total reaction force due to ligaments
 M = total reaction moment
 o_T = position vector of origin of local tibial coordinate system
 o_F = position vector of origin of local femoral coordinate
 r = position vector

Scalars:

A = cross sectional area of ligament
 F_i, M_i = total reaction force and moment components ($i = 1, 2, 3$)

l_i = direction cosines
 L_o = slack length of ligament
 L = ligament length
 L_x, L_y, L_z = length components of n th ligament
 q_i = position parameters ($i = 1, 2, 3, 4, 5, 6$)
 $Q, \bar{Q}$ = total reaction load
 T = ligament tension
 (x, y, z) = position expressed in local tibial coordinate system
 (X, Y, Z) = position expressed in local femoral coordinate system
 (X, Y, Z) = position expressed in global coordinate system
 x_i, θ_i = translation and rotation parameters ($i = 1, 2, 3$)

Superscript

n = n th ligament

Subscripts

a = point about which moments are calculated
 c = center of rotation
 F = denotes femur
 T = denotes tibia

The Model

The proposed model is for the passive structures of the knee, and therefore does not include the effects of active muscle contraction. Also, the model does not account for the viscoelastic nature of the ligaments. When the bones are displaced, the ligaments are stretched and develop forces which resist the motion. In this way, forces such as shear loads which develop during a drawer test are transmitted across the joint line. The ligamentous tensile forces also combined with the bone contact compressive forces permit transmission of moments. Since the ligaments are arranged in parallel, the forces and moments they transmit are additive. Thus the total reaction load between the tibia and femur is the sum of that due to the individual ligaments and the contact force, as shown in Fig. 1.

Figure 2, shows the coordinate system for a twelve degree of freedom equivalent beam model for the knee joint. Two types of cartesian right-hand coordinate systems are used to define the structure: a global system and two local body systems. The local body coordinate systems (x, y, z) and (X, Y, Z) are fixed in the tibia and the femur, respectively; their origin are defined by position vector $\mathbf{o}_T$ and $\mathbf{o}_F$ in the global coordinate system $(\bar{X}, \bar{Y}, \bar{Z})$.

The system of characteristic equations for the knee joint in the set of local coordinate systems (x, y, z) and (X, Y, Z) expressed in the matrix form is:

$$\begin{bmatrix} Q_{1T} \\ Q_{2T} \\ Q_{3T} \\ Q_{4T} \\ Q_{5T} \\ Q_{6T} \\ Q_{1F} \\ Q_{2F} \\ Q_{3F} \\ Q_{4F} \\ Q_{5F} \\ Q_{6F} \end{bmatrix} = \begin{bmatrix} [K_{11}] & [K_{12}] \\ [K_{21}] & [K_{22}] \end{bmatrix} \begin{bmatrix} q_{1T} \\ q_{2T} \\ q_{3T} \\ q_{4T} \\ q_{5T} \\ q_{6T} \\ q_{1F} \\ q_{2F} \\ q_{3F} \\ q_{4F} \\ q_{5F} \\ q_{6F} \end{bmatrix} \quad (1a)$$

or

$$\begin{bmatrix} Q_T \\ Q_F \end{bmatrix} = [K] \begin{bmatrix} q_T \\ q_F \end{bmatrix} \quad (1b)$$

where $[K]$ is a 12×12 stiffness matrix of the equivalent beam element modeling the knee, q_T and q_F are the displacement (translation and rotation) vectors, and Q_T and Q_F are the load (force and moment) vectors. In equations (1a and b) Q_T and q_T are given with respect to the local tibial coordinate system, while Q_F and q_F are with respect to the local femoral coordinate system. These equations can be expressed with respect to a global system $(\bar{X}, \bar{Y}, \bar{Z})$ as:

$$\begin{bmatrix} \bar{Q}_T \\ \bar{Q}_F \end{bmatrix} = [\bar{K}] \begin{bmatrix} \bar{q}_T \\ \bar{q}_F \end{bmatrix} \quad (2)$$

where $[\bar{K}]$ is the symmetric stiffness matrix of the equivalent beam element in the global system of axis. The transformation between $[K]$ and $[\bar{K}]$ given by Zienkiewicz and Cheung [18] is

$$[\bar{K}] = [\bar{T}]^{-1} [K] [\bar{T}] \quad (3)$$

where

$$[\bar{T}] = \begin{bmatrix} \Phi_T & 0 & 0 & 0 \\ 0 & \Phi_T & 0 & 0 \\ 0 & 0 & \Phi_F & 0 \\ 0 & 0 & 0 & \Phi_F \end{bmatrix} \quad (4)$$

Here $[\Phi_T]$ and $[\Phi_F]$ are the two rotation matrices which

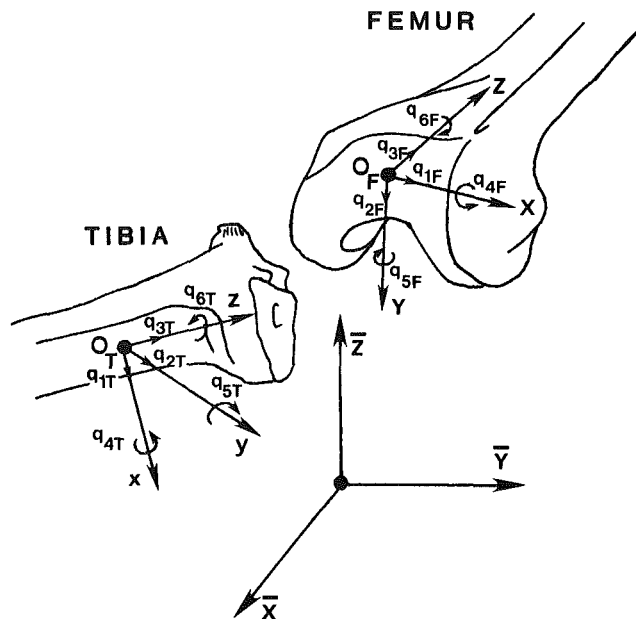


Fig. 2 Coordinate system for an equivalent beam element modeling the knee

describe the orientation of the tibial and femoral coordinate systems with respect to the global system. The vectors $[\bar{Q}_T]$, $[\bar{Q}_F]$, $[\bar{q}_T]$ and $[\bar{q}_F]$ are the tibial and femoral loads and displacements expressed in the global coordinate system. In developing our analysis, the tibia is considered to be fixed and displacements are applied to the femur. Therefore, one can rewrite equation (1) in the form:

$$\begin{bmatrix} Q_T \\ Q_F \end{bmatrix} = \begin{bmatrix} [K_{12}] \\ [K_{22}] \end{bmatrix} [q_F] \quad (5)$$

where $[K_{12}]$ and $[K_{22}]$ are 6×6 stiffness submatrices. From equilibrium we have, as given by Meek [19]:

$$[K_{22}] = -[L][\bar{R}][K_{12}] \quad (6)$$

The matrix $[\bar{R}]$ is written as:

$$[\bar{R}] = \begin{bmatrix} [R] & 0 \\ 0 & [R] \end{bmatrix} \quad (7)$$

where the matrix $[R]$ describes the relative rotational orientation between the femur and the tibia. The matrix $[L]$ is given by Meek [19] as:

$$[L] = \begin{bmatrix} [I] & 0 \\ [W] & [I] \end{bmatrix} \quad (8)$$

where $[I]$ is the unit matrix and $[W]$ is an antisymmetric matrix with off-diagonal terms describing the relative translational displacement between the tibial and the femoral origin expressed in local femoral coordinates. Therefore, equation (5) can be reduced to the following set of independent equations:

$$[Q_T] = [K_{12}][q_F] \quad (9)$$

where $[Q_T]$ are the loads applied to the tibia and $[q_F]$ are the femoral displacements with respect to the tibia.

In equation (9) the matrix $[K_{12}]$ is a secant stiffness submatrix of the knee joint. This equation can be written in the differential form if $[Q_T]$ and $[q_F]$ are replaced by their infinitesimal variations $[\delta Q]$ and $[\delta q]$, respectively, and $[K_{12}]$ is replaced by the tangent stiffness matrix $[S]$. The difference between the secant and the tangent stiffness is illustrated in

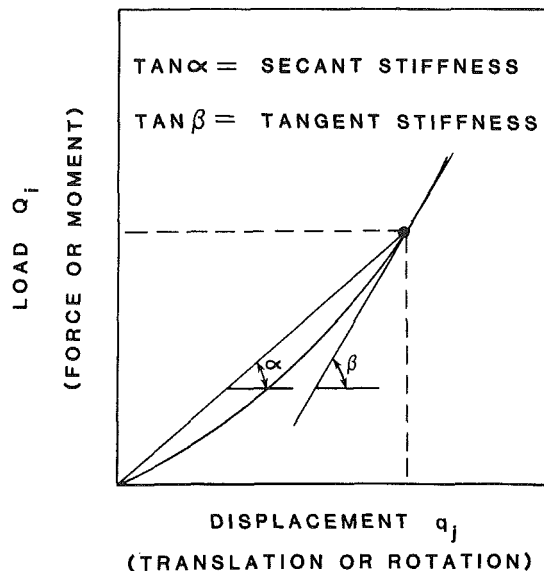


Fig. 3 Secant and tangent stiffnesses

Fig. 3. The 36 elements of the tangent stiffness matrix are given by

$$S_{ij} = \partial Q_i / \partial q_j \quad i, j = 1, 2, 3, 4, 5, 6 \quad (10)$$

For a system to be described by equation (10) the reaction load component Q_i , acting on the tibia must be a continuous differentiable function of the joint position parameters q_j . To determine the joint stiffness coefficient, S_{ij} , we first develop expressions which relate the reaction forces to the position parameters and perform the indicated differentiation.

Joint Reaction Load Due to a Single Ligament. Consider a single ligament or portion of a ligament which is denoted by the superscript, n . The tension (uniaxial force) in the ligament is given by T^n , its length by L^n , and its tibial and femoral insertion sites by x_T^n, y_T^n, z_T^n and x_F^n, y_F^n, z_F^n , respectively. Here X_F, Y_F, Z_F are femoral insertion site coordinates relative to a femoral coordinate system (cap letters) and x_T, y_T, z_T are the tibial insertion site coordinates relative to a tibial system (lower case letters). These parameters, which are required by the model, may be measured with appropriate stereotaxic equipment during careful dissection of cadaver knees.

In the analysis presented in the forthcoming, it is necessary to locate both the femoral and tibial insertion coordinates with respect to the same coordinate system. The spatial location of the femoral insertion sites, X_F, Y_F, Z_F , may be readily converted into tibial system coordinates x_F, y_F, z_F using the transformation

$$\begin{bmatrix} x_F \\ y_F \\ z_F \end{bmatrix} = [R]^T \begin{bmatrix} X_F \\ Y_F \\ Z_F \end{bmatrix} + [T] \quad (11)$$

The matrix, $[R]$, describes the relative rotational orientation between femur and tibia while the vector, $[T]$, locates the origin of the femoral system with respect to the tibial system. The superscript T indicates a transpose of a matrix.

The length of the n th ligament, L^n , is related to the location of its insertion sites, expressed relative to the tibial coordinate system, by the following equations:

$$(L^n)^2 = (x_F^n)^2 + (y_F^n)^2 + (z_F^n)^2 \quad (12)$$

where

$$\begin{aligned} L_x^n &= x_F^n - x_T^n \\ L_y^n &= y_F^n - y_T^n \\ L_z^n &= z_F^n - z_T^n \end{aligned} \quad (13)$$

These equations assume the ligament travels in a straight line between insertion sites. It is further assumed that each ligament supports only tension and has a unique relation between its tensile force, T^n , and its length, L^n , which is denoted here by $T^n(L^n)$. The time-dependent properties of creep, force relaxation, etc., are neglected, but could readily be incorporated by making the force (and therefore stiffness) depend upon strain rate and time. The relations between force and length are not known and must be experimentally determined, or otherwise specified, to permit model calculations. Since the relation is between force and length, stress-strain data from a single ligament or tendon alone are not sufficient. In addition it is necessary to know the cross-sectional area of each ligament and the length at which each ligaments first develops a force.

The line of action of the ligament force is taken to be the same as the spatial orientation of the ligament, which is specified by its direction cosines

$$l_x^n = L_x^n / L^n; l_y^n = L_y^n / L^n; l_z^n = L_z^n / L^n \quad (14)$$

If the ligament tensile force, T^n , is known from its length-tension relation, then the force due to the n th ligament F^n , acting on the tibia can be expressed in vector notation as:

$$F^n = \sum_i T^n(L^n) l_i^n e_i \quad i = 1, 2, 3 \quad (15)$$

The moment M^n about any point (a) due to the ligament force F^n is given by:

$$M^n = (r_T^n - r_a) \times F^n \quad (16)$$

where r_a is the vector locating point (a) whose coordinates are (x_a, y_a, z_a) , and r_T^n is the position vector of the insertion point of the n th ligament on the tibia. Equation (16) can be expressed as:

$$M^n = F^n \cdot \Omega(r_T^n - r_a) \quad (17)$$

where the quantity Ω is called the antisymmetric tensor of the vector, $(r_T^n - r_a)$, and has the form:

$$\Omega(r_T^n - r_a) = \begin{bmatrix} 0 & (z_T^n - z_a) & -(y_T^n - y_a) \\ -(z_T^n - z_a) & 0 & (x_T^n - x_a) \\ (y_T^n - y_a) & -(x_T^n - x_a) & 0 \end{bmatrix} \quad (18)$$

Total Joint Reaction Load. The total force F acting on the tibia due to n ligaments is determined from

$$F = \sum_n F^n = \sum_n \sum_i T^n l_i^n e_i \quad i = 1, 2, 3 \quad (19)$$

and the total moment M about any point a due to the force F is expressed as:

$$M = \sum_n F^n \cdot \Omega(r_T^n - r_a) \quad (20)$$

Stiffness Matrix

The expressions for the stiffness matrix are obtained by performing the differentiation indicated in equation (10). The components of the reaction load, Q , are the components of the forces and moments acting on the tibia, given by equations (19) and (20), respectively. The components of the position vector, q , are those which describe the translation and rotation of the femur with respect to the tibial coordinate system. Translations along the coordinate axes will be described by changes in the position parameters, q_1, q_2, q_3 or

alternately x_1 , x_2 and x_3 . Rotations about the coordinate axes will be described by changes in the parameters, q_4 , q_5 , q_6 , or alternately θ_1 , θ_2 , θ_3 .

The stiffness matrix is partitioned into four quadrants

$$[S] = \begin{bmatrix} \frac{\partial F_i}{\partial x_j} & \frac{\partial F_i}{\partial \theta_j} \\ \frac{\partial M_i}{\partial x_j} & \frac{\partial M_i}{\partial \theta_j} \end{bmatrix} = \begin{bmatrix} S1_{ij} & S2_{ij} \\ S3_{ij} & S4_{ij} \end{bmatrix} \quad (21)$$

The terms in these quadrants have the following physical significance: The change in joint reaction force acting on the tibia with respect to a translation of the femur is $\partial F/\partial x$, the change in joint reaction force acting on the tibia with respect to a rotation of the femur is $\partial F/\partial \theta$. Similarly, $\partial M/\partial x$ and $\partial M/\partial \theta$ are the changes in joint moments acting on the tibia due to a translation and rotation of the femur, respectively. It is important to note that the center of rotation, r_c , can be different than the point about which the moments are evaluated, r_a .

Since the loads applied to the tibia are vectors, they are additive; and their derivatives are additive, too. Thus, the stiffness matrix can be calculated for each of the ligaments and the resulting matrices added to obtain the stiffness matrix for the entire joint ($S = \sum_n S^n$).

For a single ligament, we have

$$F_i = T l_i \quad (22)$$

where the superscript n is dropped for convenience; using the chain rule, we get

$$\frac{\partial F_i}{\partial x_{jF}} = \frac{\partial T}{\partial x_{jF}} l_i + T \frac{\partial l_i}{\partial x_{jF}} \quad (23)$$

Since we assumed that T is a function only of ligament length, L , then

$$\frac{\partial T}{\partial x_{jF}} = \frac{dT}{dL} \cdot \frac{\partial L}{\partial x_{jF}} \quad (24)$$

Using equations (12)–(14), we obtain

$$\frac{\partial L}{\partial x_{jF}} = l_j \quad (25a)$$

$$\frac{\partial l_i}{\partial x_{jF}} = \frac{1}{L} (\delta_{ij} - l_i l_j) \quad (25b)$$

where δ_{ij} is the Kronecker-delta. Upon substituting (24) and (25) into (23), we get

$$S1_{ij} = \frac{\partial F_i}{\partial x_{jF}} = \frac{dT}{dL} l_i l_j + \frac{T}{L} (\delta_{ij} - l_i l_j) \quad (26)$$

which completes the first quadrant of S_{ij} . Note that $S1_{ij}$ is formed from two parts; the first part given by $(dT/dL) l_i l_j$ is the contribution to the knee stiffness matrix due to the ligament's axial stiffness while the second part given by $(T/L) (\delta_{ij} - l_i l_j)$ is the contribution due to a change in the ligament's orientation. These two parts correspond to the elastic and geometric portions of the stiffness matrix described by Meek [19]. From inspection, we note that both parts of $S1_{ij}$ are symmetric.

To determine the change in joint reaction forces with respect to joint rotation, we have

$$\frac{\partial F_i}{\partial \theta_{jF}} = \frac{\partial F_i}{\partial x_{kF}} \cdot \frac{\partial x_{kF}}{\partial \theta_{jF}} \quad (27)$$

where F_i is only a function of x_{kF} and x_{kT} , the latter being fixed. To determine the dependence of x_{kF} upon θ_{jF} , we start

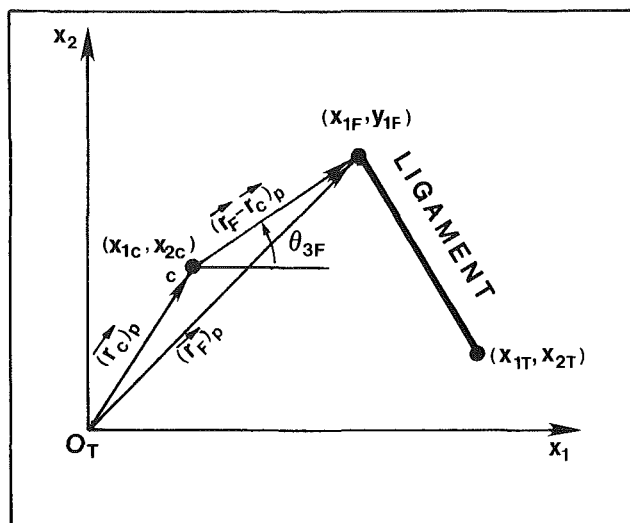


Fig. 4 The position of a femoral insertion relative to an axis of rotation parallel to the x_3 axis and passing through the point c

with a two-dimensional projection on the x_1, x_2 plane as shown in Fig. 4. The center of rotation, point c , is located by vector r_c , whose projection on the x_1, x_2 plane is $(r_c)_p$. The x_1 coordinate of the femoral insertion site is related to center of rotation by:

$$x_{1F} = x_{1c} + |(r_F - r_c)_p| \cos \theta_{3F} \quad (28)$$

By performing the differentiation with respect to θ_{3F} , for example, we obtain

$$\frac{\partial x_{1F}}{\partial \theta_{3F}} = -|(r_F - r_c)_p| \sin \theta_{3F} = -(x_{2F} - x_{2c}) \quad (29)$$

Similar expressions can be obtained for $\partial x_{iF}/\partial \theta_{jF}$. Note that $\partial x_{kF}/\partial \theta_{kF} = 0$. Arranging the results in a matrix form, we obtain

$$\frac{\partial x_{kF}}{\partial \theta_{jF}} = \Omega_{kj} (r_F - r_c) \quad (30)$$

and then substituting into equation (27), we get the following result for the upper right quadrant of the stiffness matrix:

$$S2_{ij} = S1_{ik} \Omega_{kj} (r_F - r_c) \quad (31)$$

The lower half of the stiffness matrix is obtained by differentiating the moment about point (a) , whose components are given by:

$$M_i = F_k \Omega_{ki} (r_T - r_a) \quad i = 1, 2, 3, \quad (32)$$

Note that vector r_T could be replaced with any vector along the line of action of the force. Performing the differentiation, and using the symmetry of $S1_{ij}$, we obtain

$$S3_{ij} = \Omega_{ik} (r_a - r_T) S1_{kj} \quad (33)$$

$$S4_{ij} = S3_{ij} \Omega_{kj} (r_F - r_c) \quad (34)$$

If instead of using the tibial insertion site, r_T , in equation (32) we employed the femoral insertion site, r_F then moment components become

$$M_i = F_k \Omega_{ki} (r_F - r_a) = -(x_{kF} - x_{ka}) \Omega_{ki} (F) \quad (35)$$

Differentiating equation (35) with respect to (x_{jF}) , we obtain

$$\frac{\partial M_i}{\partial x_{jF}} = -\delta_{kj} \Omega_{ki} (F) - (x_{kF} - x_{ka}) \frac{\partial \Omega_{ki} (F)}{\partial x_j} \quad (36)$$

which can be rewritten as:

$$S3_{ij} = \Omega_{ij}(\mathbf{F}) + \frac{\partial F_k}{\partial x_j} \Omega_{ki}(\mathbf{r}_F - \mathbf{r}_a) \quad (37)$$

By inspecting equation (37), the matrix $S3_{ij}$ can be written in the form:

$$S3_{ij} = S2_{ji} + \Omega_{ij}(\mathbf{F}) + S1_{jk} \Omega_{ki}(\mathbf{r}_c - \mathbf{r}_a) \quad (38)$$

Expression (38) shows that the stiffness matrix determined from our analysis is not symmetric. This conclusion is expected since this stiffness matrix corresponds to only the second quadrant in the 12×12 stiffness matrix of the equivalent beam element modeling the knee (see equations (1) and (2)).

An important result of the analysis, apparent from equations (26), (31), (33) and (34) is that, of the 36 terms in the stiffness matrix for the entire joint, only six are independent. These are the three diagonal terms of $S1_{ij}$ and the three symmetric off diagonal terms. The remaining 30 terms may be calculated from knowledge of these six independent terms, the spatial coordinates of the ligament insertion sites on the femur, $\mathbf{r}_F$, and tibia $\mathbf{r}_T$, the point where the moment is taken about $\mathbf{r}_a$ and the center of rotation is $\mathbf{r}_c$.

Also, one can easily show that the stiffness matrix S_{ij} for an arbitrary motion is evaluated from the stiffness matrix S'_{ij} , where the rotation and the moments are taken about the origin, as:

$$S1_{ij} = S1'_{ij} \quad (39a)$$

$$S2_{ij} = S2'_{ij} - S1'_{ik} \Omega_{kj}(\mathbf{r}_c) \quad (39b)$$

$$S3_{ij} = S3'_{ij} + \Omega_{ik}(\mathbf{r}_a) S1'_{kj} \quad (39c)$$

$$S4_{ij} = S4'_{ij} - S3'_{ik} \Omega_{kj}(\mathbf{r}_c) + \Omega_{ik}(\mathbf{r}_a) S2'_{kj} - \Omega_{im}(\mathbf{r}_a) S1'_{mk} \Omega_{kj}(\mathbf{r}_c) \quad (39d)$$

Model Calculations

Since the primary objective of this paper is to present an analytical technique, only a limited number of calculations have been conducted. The purpose of these calculations was to compare our model predictions with those of Crowninshield, et al. [1] using their input data for the ligaments, and their assumed joint motion. Crowninshield et al. modeled the ligaments and capsule of the knee with 13 elements. The

coordinates of their attachment sites, and their cross-sectional areas are given in their Table 1. The equations governing the motion of the knee which are used to calculate the current femoral insertion coordinates of each ligament are given in pp. 398-399 in Crowninshield, et al. [1]. The relation we used between tensile force and ligament length was derived from the data presented by Crowninshield, et al. as:

$$T = 750A \left[\frac{L - L_o}{L_o} \right]^2 \quad (40)$$

where A and L_o are the cross-sectional area and the slack length of the ligament, respectively. The slack length, L_o , of the ligaments is determined by satisfying Crowninshield, et al. [1] assumption, that the maximum length of each ligament during the motion corresponds to an extension ratio of 1.05. The extension ratio for the 13 ligament elements at full extension are presented in Table 1, where we followed the same format used by Crowninshield, et al. [1] for ligament identification.

Although it may appear that our analysis involves complicated and lengthy computations and computer programming, the expressions describing the stiffness of the knee joint given by equations (26), (31), (33) and (34) are in fact very easy to program because they are written in matrix notation. Only very few lines of Fortran code are required

Table 1 Extension ratios of the ligament elements at full extension

Ligament elements	Extension ratio
Lateral collateral	1.050
Posterior fibers of the medial collateral	1.050
Anterior fibers of the medial collateral	0.996
Oblique medial collateral	1.031
Deep medial collateral	1.049
Anterior fibers of the posterior cruciate	1.050
Posterior fibers of the posterior cruciate	1.050
Anterior fibers of the anterior cruciate	1.031
Posterior fibers of the anterior cruciate	1.012
Lateral fibers of the posterior capsule	1.050
Medial fibers of the posterior capsule	1.050
Oblique posterior capsule no. 1	1.050
Oblique posterior capsule no. 2	1.042

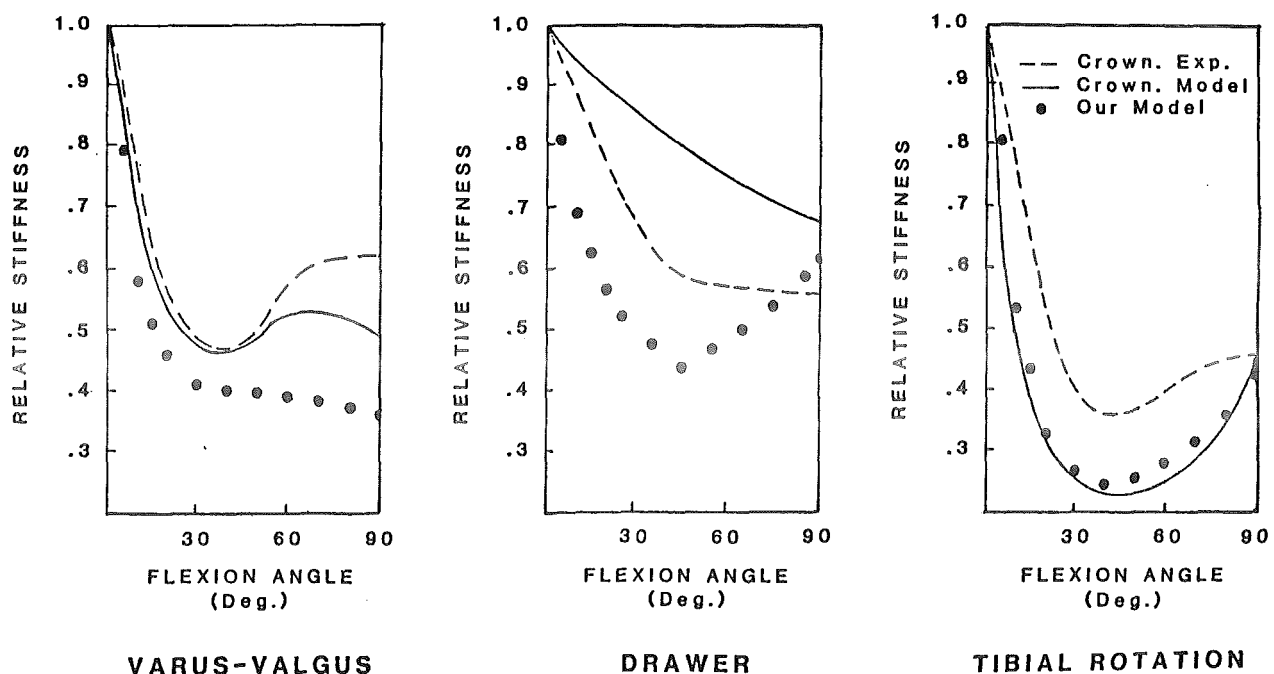


Fig. 5 Comparison of theoretical and experimental knee stability

which were verified by comparison with manual computations. The remaining part of the computer program is required for bookkeeping of each ligament's insertion site coordinates and for calculation of ligament length. The ligament length patterns we calculated were found to be identical to those presented by Crowninshield, et al. [1].

Comparisons of predictions of joint stiffness from both models with Crowninshield's, et al. average experimental data are shown in Fig. 5. Both models predict the same behavior for the relative torsional stiffness ($\partial M_3 / \partial \theta_3$) term where the moment is taken about the center of rotation $\mathbf{r}_c = \mathbf{r}_a = (-15, 0, 0)$. Some differences exist in both anterior drawer stiffness ($\partial F_2 / \partial X_2$) and varus-valgus stiffness ($\partial M_2 / \partial \theta_2$), which is based on a moment taken about the two centers of rotation $\mathbf{r}_c = \mathbf{r}_a = (-25, 0, 0)$ and $\mathbf{r}_c = \mathbf{r}_a = (25, 0, 0)$, respectively.

Differences between our results and those of Crowninshield, et al. are expected because our values are from calculations at a single joint position, while their stiffnesses are based upon a finite joint displacement and calculated from the approximate equation:

$$S_{ij} \approx \Delta Q_i / \Delta q_j \quad (41)$$

Therefore, the computations presented here primarily serve as a demonstration of the equations and the technique. The validity and accuracy of our model cannot be judged from the comparison shown in Fig. 5 because the results of our calculations depend strongly upon the ligament properties and joint motion, which Crowninshield, et al. assumed, and which we used for input data.

Concluding Remarks

We presented a rather straightforward approach for modeling the contribution of the ligaments to the nonlinear, coupled stiffness characteristics of the human knee. The potential advantage of our approach is the ability to determine stiffness directly from the problem geometry and the constitutive equations of the ligaments. A further advantage is the use of matrix notation which permits easy and compact coding of the equations. In contrast, Crowninshield's model required stiffness be obtained from a simulated joint displacement, where it is calculated from the approximate relation given by (41). Piziali [10] experimentally determined the coefficients of this matrix for a single left human knee as a family of continuous curves each related to a specific initial state of displacements and rotations. To utilize his experimental data in determining the stiffness matrix for a given joint position not tested experimentally, one must use an interpolation scheme in six space. This requires a large number of curves which must geometrically bound all anticipated positions of the knee and provide sufficient points for accurate interpolation.

The model presented here permits calculation of the tangent stiffness $[S]$ of the knee joint directly from problem geometry and the constitutive equations of the ligaments. Accordingly, the joint reaction forces and moments due to the ligaments are related to the joint position parameters, q_j , by a set of integrals having the form:

$$Q_i = \int_{\text{initial}}^{\text{final}} S_{ij} dq_j \quad i, j = 1, 2, 3, 4, 5, 6 \quad (42)$$

The model is useful in solving for the static equilibrium position of the tibia relative to the femur when bones are subjected to arbitrarily specified external forces. These will include gravitational loads, internal forces of the muscles, joint contact forces, and the external forces and moments such as those due to activity or those imposed by a physician in performing a clinical evaluation of laxity.

In the formulation of the equations, one of the bones is

taken rigidly fixed in space and equilibrium between the femur and the tibia is determined by imposing the laws of statics on the remaining bone. The 12 variables describing the problem are the applied forces and moments Q_i^* , $i = 1, 2, 3, 4, 5, 6$, and the joints translations and rotations, q_j , $j = 1, 2, 3, 4, 5, 6$. The six equations of equilibrium are:

$$Q_i^* + \int_{\text{initial}}^{\text{final}} S_{ij} dq_j + Q_{ci} = 0 \quad i, j = 1, 2, 3, 4, 5, 6 \quad (43)$$

where Q_{ci} are the components of the contact forces between the femoral and tibial condyles. Depending on the joint position, there are at most two contact forces which are utilized when solving these six equations of equilibrium to determine the six unknown variables when the remaining six are given.

In the displacement version of the model, the desired joint displacements, q_j , are specified and the resulting reaction loads, Q_i , are calculated directly from equations (12) through (16) without the need for integration. Here, only the ratios between the contact forces, if any, and the applied loads can be determined using the equations of equilibrium.

In the second mode, the forces and moments applied to the tibia are specified as a function of time. A repetitive solution algorithm with an appropriate convergence criteria is used to determine the unknown displacements. These numerical techniques can also be used in the mixed version of the model where any combination of six variables are given and the other six are determined.

We believe that the modeling approach presented here is most useful in the solution of more complex problems where the joint equilibrium position needs to be determined when the external loads and internal muscle forces are known. In the second part of this work [20], the authors will present a model of the ligamentous geometric nonlinearities due to wrapping of the ligaments around each other (cruciates) and around bone surfaces (posterior capsule, medial collateral, and cruciates at full extension).

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Errata¹

APPENDIX Matrix Formulation of the Problem

$$\begin{bmatrix} (1+A) & -E_1 & 0 & 0 & \dots & 0 & 0 & 0 \\ -\frac{p}{2} & (1+c_1) & -\frac{p}{2} & 0 & \dots & 0 & 0 & 0 \\ 0 & -\frac{p}{2} & (1+c_2) & -\frac{p}{2} & \dots & 0 & 0 & 0 \\ . & . & . & . & \dots & . & . & . \\ . & . & . & . & \dots & . & . & . \\ 0 & 0 & 0 & 0 & \dots & -\frac{p}{2} & (1+c_{M-1}) & -\frac{p}{2} \\ 0 & 0 & 0 & 0 & \dots & 0 & -E_{M-1} & (1-D) \end{bmatrix} \begin{bmatrix} \theta_0^{n+1} \\ \theta_1^{n+1} \\ \theta_2^{n+1} \\ . \\ . \\ \theta_{M-1}^{n+1} \\ \theta_M^{n+1} \end{bmatrix} +$$

$$\begin{bmatrix} (1-A) & E_1 & 0 & 0 & \dots & 0 & 0 & 0 \\ -\frac{p}{2} & (1-c_1) & \frac{p}{2} & 0 & \dots & 0 & 0 & 0 \\ 0 & \frac{p}{2} & (1-c_2) & \frac{p}{2} & \dots & 0 & 0 & 0 \\ . & . & . & . & \dots & . & . & . \\ . & . & . & . & \dots & . & . & . \\ 0 & 0 & 0 & 0 & \dots & \frac{p}{2} & (1+c_{M-1}) & \frac{p}{2} \\ 0 & 0 & 0 & 0 & \dots & 0 & E_{M-1} & (1-D) \end{bmatrix} \begin{bmatrix} \theta_0^n \\ \theta_1^n \\ \theta_2^n \\ . \\ . \\ \theta_{M-1}^n \\ \theta_M^n \end{bmatrix} +$$

$$2pB\Delta\xi \begin{bmatrix} 1 \\ 0 \\ 0 \\ . \\ . \\ 0 \\ 0 \end{bmatrix} = 0$$

where

$$A = p \left\{ 1 + B\Delta\xi + \frac{3\beta_0(\Delta\xi)^2}{8} \right\} \quad (15)$$

$$c_m = \left\{ 1 + \frac{\beta_m(\Delta\xi)^2}{2} \right\} \quad (16)$$

$$D = p \left\{ 1 + \frac{3\beta_M(\Delta\xi)^2}{8} \right\} \quad (17)$$

$$E_m = p \left\{ 1 - \frac{\beta_m(\Delta\xi)^2}{8} \right\} \quad (18)$$

$$p = \Delta\tau / (\Delta\xi)^2 \quad (19)$$

$n, n+1$ represent time steps such that

$$\tau_{n+1} - \tau_n = \Delta\tau \quad (20)$$

¹For "Transient Temperature Profiles in Tissues With Nonuniform Blood Flow Distributions," by A. B. Elkowitz, A. Shitzer, and R. C. Eberhart, published in the August 1982 issue of the *JOURNAL OF BIOMECHANICAL ENGINEERING*, Vol. 104, pp. 202-208.